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Xylem anatomy differentiation explains coordinated variation of economic and hydraulic traits in urban tree species

Shuzhen Lv^{1†}, Yanqiong Meng^{1,2†}, Nuo Huang¹, Liuting Li¹, Jianan Wang^{1,2} and Yiyong Li^{1,2*}

Abstract

Background Urban drought increasingly challenges the survival and performance of urban forests, consequently undermining their sustainability and productivity. Hydraulic-economic trait trade-offs are primary determinants of species drought tolerance, yet these trade-offs remain poorly understood in urban forests.

Methods In this study, we investigated the differentiation in hydraulic and economic traits in leaf and stem functional traits among 14 tree species with different xylem anatomical types, explored their correlations along the plant economic spectrum, and further systematically evaluated drought adaptation strategies, aiming to guide species selection and management in urban forests.

Results We found that ring-porous trees exhibited trait values associated with faster water transport compared to diffuse-porous species. Specifically, ring-porous species had significantly higher xylem-specific conductivity, leaf-specific conductivity, and branch xylem vessel diameter. However, the hydraulic safety margin of diffuse-porous species was significantly lower than that of ring-porous species, accompanied by lower leaf and wood phosphorus content. These patterns support the hydraulic safety enhancement strategy in ring-porous species. Across the 14 studied tree species, ring-porous species occupied the fast-acquisitive end of the plant economics spectrum, while diffuse-porous species were positioned at the conservative end. We also discovered a significant negative correlation between hydraulic and economic functional trait principal components, demonstrating that hydraulic safety and carbon economy are synergistically integrated along a unified plant economics spectrum, from fast-growing but vulnerable species to slow-growing but resilient ones.

Conclusions Our study reveals that ring-porous and diffuse-porous species occupy opposite ends of the plant economics spectrum, employing divergent drought adaptation strategies. This clear functional differentiation provides a robust ecological rationale for diversifying urban forests. We propose that selecting a mixture of acquisitive and conservative species will create a more resilient urban ecosystem capable of withstanding a variety of environmental stresses, from seasonal droughts to heatwaves, thereby securing long-term stability and functionality.

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Keywords Functional traits, Hydraulic traits, Xylem hydraulics, Economics spectrum, Stoichiometry, Urban forest, Urbanization

Introduction

Climate change-related drought risks are intensifying globally [1, 2], which systemically impair plant development and physiological processes, compromising primary productivity and nutrient cycling efficiency [3–6]. In many urban areas, the drought effect is expected to be amplified by urban heat island (UHI) effect, thereby resulting in more severe environmental stress in cities, especially at highly urbanized area [7]. Global climate change-associated droughts are resulting in the occurrence of increasing widespread tree mortality, as highlighted by considerable evidences [8, 9]. These effects are posing a significant threat to the ecological benefits of urban vegetations, especially in water and carbon cycling processes. Compared to natural environments, stressors such as the urban heat island effect and atmospheric pollution disrupt normal hydraulic and carbon investment by forcing stomatal closure and impairing photosynthesis, thereby restricting carbon uptake and disturbing carbon-water balance [10, 11]. Previous studies suggest that under climate warming scenarios, the survival capacity of urban plants is determined by the adaptability of their hydraulic systems to extreme drought [12, 13]. Even within the same region, different tree species facing the same drought stress may adopt divergent adaptive strategies [14]. Therefore, investigating the differences and coordination in hydraulic architecture and resource acquisition strategies is critical for selecting adaptive tree species and for the effective management of urban forestry, as well as for enhancing ecosystem resilience under conditions of increasing drought.

The evolutionary divergence between diffuse-porous (DP), semi-ring-porous (SRP), and ring-porous (RP) woods represents a paradigmatic case of xylem adaptation to contrasting environmental regimes [15, 16]. Anatomically, RP species develop distinguishable growth rings with earlywood vessels exceeding 200 μm in diameter (e.g., *Quercus* spp.), whereas DP species exhibit radially uniform vessel distributions with diameters typically below 100 μm (e.g., *Acer* spp.). Accordingly, the vessel diameter of SRP species typically falls between that of RP and DP woods [17–19]. RP species large-vessel porosity fulfills rapid water transport demands during spring growth triggered by snowmelt [20, 21]. Conversely, DP species employs dual strategies: homogeneous vessel distribution ensures year-round hydraulic stability in tropical rainforests by mitigating environmental variability, while minimized vessel diameter variation reduces freeze-thaw embolism risks in cold climates through structural uniformity [22]. These strategies are reflected

in physiological and morphological features that indicate differences among species in terms of their adaptive design. For example, wood hydraulic conductivity and leaf-specific hydraulic conductivity are considered to reflect a range of hydraulic functionalities [23, 24], with high vessel density and high hydraulic safety margin being linked to increased hydraulic safety and an associated capacity for maintaining operational functionality at lower water potentials [25, 26]. The two groups exhibit disparities not merely in wood structure, but also in hydraulic strategies and resource allocation trade-offs under environmental stress [27, 28]. For example, RP species exhibit superior hydraulic efficiency, photosynthetic rates, and stem biomass increments, enabling rapid resource acquisition and growth, whereas DP species adapt to high-altitude harsh environments through conservative traits such as extended leaf lifespan [29]. These growth-survival trade-offs are underpinned by divergent elemental investment strategies: an N- and P-oriented strategy facilitates rapid tissue construction, whereas a C-based conservative strategy enhances structural durability and resilience, thereby establishing these traits as fundamental dimensions of the plant economic spectrum [30–32]. Emerging evidence suggests critical divergences in drought adaptation strategies between RP and DP species [33, 34]. Bryant et al. [35] demonstrated contrasting hydraulic responses to water stress between DP *Acer saccharum* and RP *Carya ovata* in temperate deciduous forests by comparing their growing-season hydraulic behaviors. The DP species maintained stable leaf water potential through rapid stomatal closure, prioritizing avoidance of hydraulic failure, while the RP species sustained photosynthesis via delayed stomatal regulation at the cost of lower leaf water potentials and elevated hydraulic safety risks, a strategy that demands greater investment of nitrogen and phosphorus for the repair of tissues and synthesis of osmotic compounds [36–38]. Although functional traits of RP and DP species have been extensively compared across various environmental contexts and scales, hydraulic-resource strategy divergence among wood anatomical groups—such as semi-ring-porous species—has received far less attention, particularly under urban drought conditions.

As key organs integrated within the plant hydraulic continuum, stems and leaves exhibit hydraulic-economic coordination (coupling vs. decoupling) that critically governs whole-plant drought resilience [39–41]. By enhancing water uptake and transport, higher shoot hydraulic conductivity helps maintain leaf-level hydraulic conductance and instantaneous carbon assimilation [42, 43],

further supporting elevated photosynthetic rates that enable ring-porous species to accumulate more photoassimilates and achieve greater growth [44]. Emerging evidence revealed organ-level coordination among woody plants: Zhou et al. (2022) working across 61 temperate tree species, documented interspecifically coupled acquisitive strategies in stem-leaf organs of temperate woody plants [45], whereas a study of 18 mangrove species [46], observed interspecific decoupling between stem hydraulic traits and leaf physiological traits (e.g., stomatal-vein density coordination) compared to global datasets. Under water stress, possible mechanisms underpinning this decoupling are thought to arise from species-specific hydraulic segmentation via leaf abscission or stomatal closure, prioritizing stem hydraulic integrity at the expense of carbon assimilation [47, 48]. Meanwhile, it has been suggested that coupling mechanisms—whereby woody plants enhance stem-to-leaf water subsidies through tissue capacitance and osmotic adjustment—sustain physiological activity under fluctuating resource availability [49]. However, how this coordination adapts to unique urban environmental stressors remains poorly understood. Therefore, in this study, we measured a suite of key hydraulic and economic traits in stems and leaves (e.g., xylem-specific conductivity, leaf carbon content) to assess the implications for urban tree resilience and functionality under anthropogenic pressures to fundamentally elucidate how xylem anatomical differentiation drives holistic adaptation strategies to drought and multiple abiotic stresses in woody plants, particularly within urban environments.

Here, we assessed how functional traits describing hydraulic attributes and the economic spectrum vary among three wood anatomical types and how they

coordinate to shape ecological strategies. We studied this by conducting anatomical experiments on branches and leaves of 14 woody plant species in Hefei urban areas, categorizing them into DP, SRP, and RP types. In particular, we addressed the following questions: (1) Do the differences in hydraulic and economic traits among urban tree species belonging to different wood anatomical types become smaller under hotter and drier urban environmental conditions? (2) Do urban tree species exhibit significant hydraulic safety-efficiency trade-offs and coordinated economic traits between stem and leaf organs? Overall, we expect differences in hydraulic functional traits between branches and leaves, as well as a tight coordination between hydraulic functional traits and economic traits, which together shape strategies for tree growth. In urban environments which often have heat and drought stress, we expect trees to have close association and interaction of hydraulic and economic traits to effectively enhance overall plant performance.

Material and method

Study site and plant sampling

The study was carried out at Luzhou Park (31°53′ N, 117°13′ E), located in Luyang District, Hefei City, Anhui Province, China. The study site is characterized by a representative northern subtropical monsoon climate. The annual precipitation is 900–1100 mm, and approximately 40% of this is concentrated in the growing season (June–August), with droughts being frequent. The average growing-season temperature ranges from 28 to 30 °C, and extreme high temperatures exceed 39 °C. Over recent decades, rapid urban expansion in Hefei City has triggered thermal fluctuations that significantly disrupt local ecosystems and intensify water stress on urban trees [40, 41].

The study includes 14 species with six individuals per species (84 total). Sampling was performed in August 2023 at four DP species (*Bischofia polycarpa*, *Liriodendron chinense*, *Acer buergerianum*, *Prunus serrulata* var. *Lannesiana*), four SRP species (*Ligustrum lucidum*, *Diospyros kaki*, *Triadica sebifera*, *Cinnamomum camphora*) with two evergreens (*Ligustrum lucidum*, *Cinnamomum camphora*), and six RP species (*Sophora japonica*, *Cotinus coggygria*, *Pistacia chinensis*, *Zelkova serrata*, *Celtis sinensis*, *Sapindus saponaria*). The remaining twelve species were deciduous (Table 1). For each species, six adult individuals of similar diameters were selected. To ensure methodological validity, we sampled healthy, two-year-old branches with fully expanded and undamaged leaves from mature, planted trees located in low-disturbance zones within the park, situated away from roadways and human activity. Furthermore, we confined our sampling to areas with similar slope, aspect, and elevation to minimize the variation in key environmental factors such as

Table 1 List of tree species selected in this study

Species	Family	Wood type	Leaf habit
<i>Bischofia polycarpa</i>	Phyllanthaceae	Diffuse-porous	Deciduous
<i>Liriodendron chinense</i>	Magnoliaceae	Diffuse-porous	Deciduous
<i>Acer buergerianum</i>	Sapindaceae	Diffuse-porous	Deciduous
<i>Prunus serrulata</i> var. <i>lannesiana</i>	Rosaceae	Diffuse-porous	Deciduous
<i>Ligustrum lucidum</i>	Oleaceae	Semi-ring-porous	Evergreen
<i>Diospyros kaki</i>	Ebenaceae	Semi-ring-porous	Deciduous
<i>Triadica sebifera</i>	Euphorbiaceae	Semi-ring-porous	Deciduous
<i>Cinnamomum camphora</i>	Lauraceae	Semi-ring-porous	Evergreen
<i>Sophora japonica</i>	Fabaceae	Ring-porous	Deciduous
<i>Cotinus coggygria</i>	Anacardiaceae	Ring-porous	Deciduous
<i>Pistacia chinensis</i>	Anacardiaceae	Ring-porous	Deciduous
<i>Zelkova serrata</i>	Ulmaceae	Ring-porous	Deciduous
<i>Celtis sinensis</i>	Cannabaceae	Ring-porous	Deciduous
<i>Sapindus saponaria</i>	Sapindaceae	Ring-porous	Deciduous

soil type and moisture availability. Sampling was conducted during the peak growing season to ensure physiological activity. For each individual, three south-facing branches from the sun-exposed upper canopy were sampled before dawn. Specifically, we selected branches that could provide a continuous unbranched stem segment of ~10 cm length and 6–10 mm diameter. Two additional branches 0.5–1.5 m long, each bearing 20–40 leaves, were collected for leaf hydraulic trait measurements. All branches were kept wet and transport to the laboratory immediately.

Stem hydraulic conductance measurement

A xylem embolism meter (XYL 'EM-Plus, Bronkhorst, France) was used to measure xylem hydraulic conductance. Before measurements, a roughly 10 cm long stem base segment was cut underwater from pre-collected samples, wrapped in black plastic bags, and allowed to rehydrate overnight. They were then flushed with degassed 0.02 mol L⁻¹ of KCl solution at 0.15 MPa for ≥20 min under laboratory conditions (20–25°C) until complete bubble removal from samples, aiming to eliminate native xylem embolism. Subsequently, four measurements at ~0.5 MPa pressure yielded mean maximum hydraulic conductance ($K_{\max}$). After measurements, stem segments were measured with digital calipers to calculate sapwood area (AS, mm²). All distal leaves from sampled branches were scanned using a scanner (BenQ K810, BenQ, China) to determine total leaf area (AL, mm²). Xylem-specific conductivity (K_s , kg m⁻¹ s⁻¹ MPa⁻¹) was calculated as K_h/AS , while leaf-specific conductivity (K_L , kg m⁻¹ s⁻¹ MPa⁻¹) was derived from K_h/AL .

Leaf hydraulic traits measurement

The Water potential at the turgor loss point (Ψ_{TLP} , -MPa) was determined by establishing pressure-volume curves (P-V). After rehydrating the branches for 2 h, we cut 3 to 6 leaves from each branch to measure the P-V relationship. During slow dehydration, leaf water potential (Ψ_{leaf} , -MPa) was periodically measured with a pressure chamber (PMS1000, Corvallis, USA) and leaf mass with an analytical balance (CPA225D, Sartorius Inc., Germany). By plotting the P-V curve with the relative water deficit (1-relative water content) as the x-axis and the inverse of the leaf water potential as the y-axis, the Ψ_{TLP} was identified as the inflection point on this curve where the relationship becomes linear, indicating the transition from turgor loss to purely elastic cell wall shrinkage.

To quantify leaf hydraulic conductivity (K , mmol m⁻² MPa⁻¹ s⁻¹), we harvested new branches the next day and dehydrated them in darkness for 30–60 min to allow for full water potential equilibrium. After this period, when leaf water potential reached -1 to -2 MPa, we measured initial water potential (Ψ_0) in two adjacent leaves. This

specific hydration range was selected based on pilot studies and species-specific vulnerability curves to ensure it remained well above the documented embolism onset point (P12) for all species in this study, thereby minimizing the risk of inducing significant embolism that could affect the determination of maximum conductivity. If these showed similar Ψ_0 values, additional leaves with comparable water potential were selected from the same branch. Selected leaves were then immersed in water, and their petioles/midribs were cut underwater. Following 15–20 s rehydration (duration adjusted by initial Ψ_0), we recorded the post-rehydration water potential (Ψ_1) at half the initial Ψ_0 value. The K was calculated using the formula: $K = C_{leaf}/T \times \ln(\Psi_0/\Psi_1)$, where C_{leaf} is the leaf capacitance (mmol m⁻² MPa⁻¹), T is the time interval (s) over which water flow was measured, and Ψ_0 and Ψ_1 are the initial and final leaf water potentials (MPa), respectively.

During a 3–4 day natural drying period, leaf images were captured every 5 min using a custom-built imaging device. The imaging area covered ≥300 mm² of leaf surface, including the midrib and veins. Concurrently, Ψ_{leaf} was measured every 1–3 h between 07:00 and 19:00 using a pressure chamber. At each measurement time point, 2–3 leaves were sampled to ensure water potential equilibrium. Laboratory temperature was maintained at 22 °C throughout the experiment to minimize the influence of temperature on the kinematic viscosity of water and on leaf metabolic processes, ensured data comparability, and kept leaf temperature fluctuations within 1 °C. Image sequences were analyzed using ImageJ software to quantify occluded pixels per image. For each tree across all 14 species, a leaf hydraulic vulnerability curve was generated by plotting the cumulative percentage of occluded pixels (relative to the final maximum occlusion in the sequence) against the corresponding Ψ_{leaf} , with successful measurements obtained for 13 species following the exclusion of *Ligustrum lucidum* due to methodological constraints. Complete embolism was confirmed by dehydrating a subset of leaves until no further increase in pixel occlusion was observed, typically at water potentials more negative than -4.0 MPa. The hydraulic safety margin (HSM, -MPa) was quantified as the difference between the leaf water potential at 50% loss of hydraulic conductivity (P50, -MPa) and the leaf water potential at 12% loss of hydraulic conductivity (P12, -MPa), which is defined as the water potential at the inception of embolism [50].

Leaf photosynthetic and morphological trait measurements

Photosynthetic traits were measured using the photosynthetic fluorescence meter Li-6800 (Lincoln, USA). Sun-exposed branches were excised and their bases

were immediately recut underwater to maintain hydraulic continuity. Photosynthetic measurements were then conducted on leaves that remained attached to these branches, performed at the field site within minutes of excision. To ensure that the photosynthetic rates of all tree species were measured under consistent conditions, all measurements were conducted between 9:00 and 11:30. Furthermore, the same environmental parameters (e.g., photosynthetically active radiation at $1200 \mu\text{mol m}^{-2} \text{s}^{-1}$, CO_2 concentration at 400 ppm, leaf temperature at 30°C , and a relative humidity of 60%) were set in the photosynthetic tester before the measurement. The area-based photosynthetic rate (A_{area} , $\mu\text{mol m}^{-2} \text{s}^{-1}$) was recorded directly. Subsequently, the mass-based photosynthetic rate (A_{mass} , $\mu\text{mol g}^{-1} \text{s}^{-1}$) was derived through conversion using leaf mass per area (LMA, g m^{-2}), calculated as $A_{\text{mass}} = A_{\text{area}}/\text{LMA}$.

Measurements of leaf thickness (LT, mm) and LMA were exclusively conducted on sun-exposed leaves. Each leaf or leaflet was scanned to calculate its area (HP Scanjet G3110, Hewlett-Packard Development Co., China). After oven-drying at 65°C for 72 h, leaf dry mass was measured. LMA was calculated as the ratio of leaf dry mass to fresh leaf area.

Stem anatomical traits measurement

Stem transverse Sect. ($20 \mu\text{m}$ thickness) were prepared using a sliding microtome (Model 2010-17, Shanghai Medical Instrument Corp., China), stained with 0.1% toluidine blue, and imaged at $10\times$ and $40\times$ magnification under a light microscope (Leica ICC50, Wetzlar, Germany) with an integrated digital camera. Wood porosity types (RP, DP, or SRP) were classified for each species based on qualitative assessment of the vessel distribution and the variation in vessel diameter across the growth ring (Fig. 1), following standard wood anatomical identification criteria [50]. Wood anatomical traits were quantified from the digital micrographs using Adobe Photoshop 25.5.0. All measurements were conducted within the mature secondary xylem of the two-year-old branches, avoiding the vicinity of the vascular cambium and the pith. Vessel density (VD) was quantified by analyzing a minimum of 15 images per stem sample taken at $10\times$ magnification, assuming circular vessel geometry. The stem xylem vessel diameter (D_{stem}) was calculated as: $D_{\text{stem}} = (\text{Rd } 4/n)^{1/4}$, where d is individual conduit diameter and n is the number of conduits. To ensure statistical robustness, D_{stem} was measured from at least 200 vessels per stem sample, which were sampled from multiple $40\times$ magnification images.

Stoichiometric determination

After collection, all leaf and stem samples were immediately placed into labeled paper envelopes and stored

in black plastic bags during transportation to the laboratory. In the lab, the stem and leaf samples were dried at 65°C for 72 h, ground using a ball mill, and passed through a 100-mesh sieve to ensure homogeneity. For subsequent elemental analysis, a representative aliquot of approximately 5 g of each powdered sample. The leaf carbon content (LCC), leaf phosphorus content (LPC), leaf nitrogen content (LNC), wood carbon content (WCC), wood nitrogen content (WNC), and wood phosphorus content (WPC) were then measured. Among them, the carbon content and nitrogen content was determined by a completely automated elemental analyzer (Vario EL cube, Elementar, Germany). The phosphorus content was determined by Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES, Leeman, USA). Subsequently, the wood carbon to nitrogen ratio (WCN) and leaf carbon to nitrogen ratio (LCN) were calculated.

Statistical analysis

Statistical analysis was conducted using the R version 4.4.2. Differences in all measured leaf hydraulic and economic traits across timber types were analyzed using linear mixed models (LMMs), with “wood type” as a fixed effect and “species” as a random effect nested within timber type groups. When significant differences were detected, a Tukey’s Honestly Significant Difference (HSD) post hoc test was subsequently applied to identify specific variations among groups. The Sigmoidal function from the *fitplc* package was employed to fit the leaf vulnerability curves [51]. We employed principal component analysis (PCA) with the “vegan” package to characterize the resource-economic spectrum coordination model across both organ and whole-plant levels. The positions of species along the first two principal components (PC1 and PC2) were used to assess differences among the three timber types via t-tests. The relationship between plant economic traits and hydraulic traits was systematically evaluated using Standardized Major Axis (SMA) regression. Pearson correlation analysis ($p < 0.05$) was used to test the correlation between individual parameters.

Result

Variation in stem and leaf functional traits among xylem anatomy types

The results showed that RP wood exhibited significantly greater K_s , K_L , D_{stem} , and WPC than DP wood (Fig. 2; $P < 0.05$). SRP species showed no significant differences in these traits relative to either group. Conversely, DP wood had significantly higher VD and WCN compared to RP wood, with VD in DP wood being significantly higher than that in SRP wood (Fig. 2; $P < 0.05$). However, no significant differences were observed in WCC and WNC.

In the analysis of leaf differences, the results showed that the HSM of leaves in RP wood was significantly

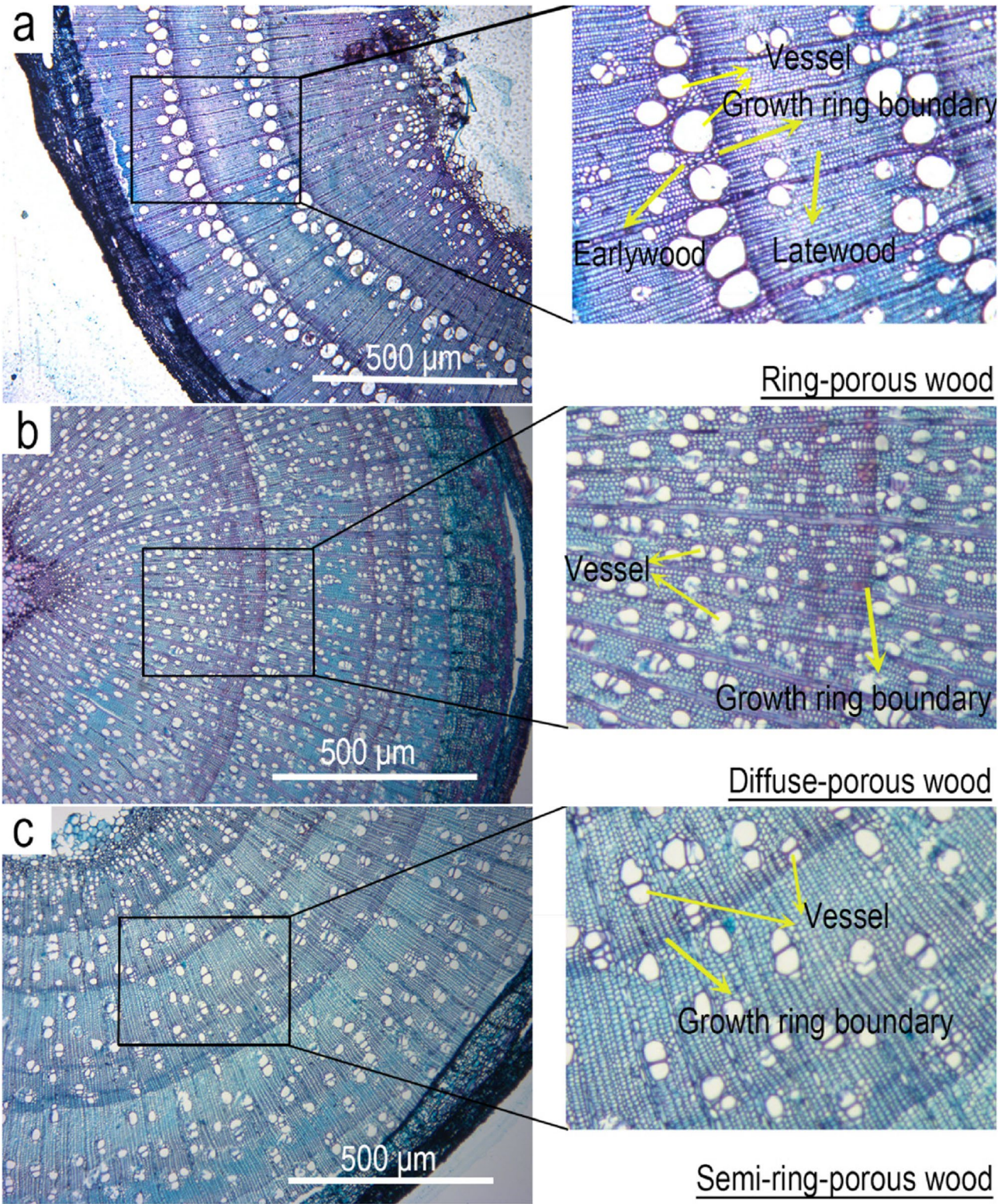


Fig. 1 Transverse sections of wood illustrating the three xylem porosity types. **a** RP wood, characterized by large-diameter vessels in the earlywood forming a distinct ring. **b** DP wood, with vessels of relatively small and uniform diameter distributed evenly throughout the growth ring. **c** SRP wood, showing a gradual transition from larger to smaller vessel diameters from earlywood to latewood, representing an intermediate state

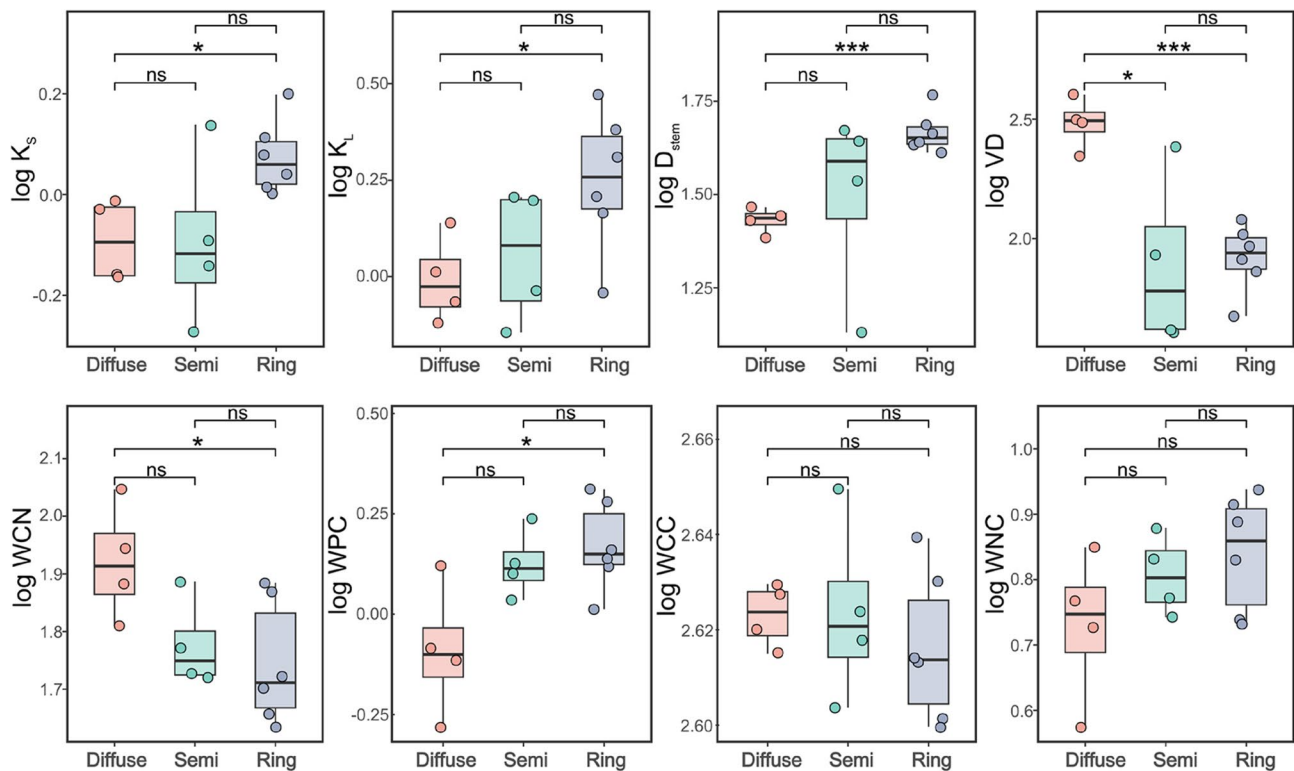


Fig. 2 Effect of urbanization on stem functional traits. K_s : xylem-specific conductivity, K_l : leaf-specific conductivity, D_{stem} : stem xylem vessel diameter, VD : vessel density, WCN : wood carbon to nitrogen ratio, WPC : wood phosphorus content, WCC : wood carbon content, WNC : wood nitrogen content. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, ns = non-significant

greater than that in DP wood (Fig. 3; $P < 0.05$). Conversely, DP wood had significantly higher LT and LCC compared to RP wood, with LPC in RP wood being significantly higher than that in SRP wood. In addition, the three wood types showed no significant differences in leaf water potential at the K, Ψ_{TLP} , Amass, LMA, LCN, and LNC (Fig. 3).

Trait coordination across xylem anatomy types

PCA of stem traits showed that the first two principal components together explained 70.01% of the total variance (Fig. 4a). The PC1 was represented by stem acquisition strategy-related traits including K_s , D_{stem} , K_l , WPC, and WNC, all positioned on the right side of the PC1 axis and gradually increasing, while WCC, WCN, and VD were positioned on the left side of the PC1 axis and gradually decreasing, explaining 51.70% of the total variance. PCA of leaf traits showed that the first two principal components together explained 58.85% of the total variance (Fig. 4b). The PC1 was represented by leaf conservative strategy-related traits including LCC, LCN, LMA, LT, and Ψ_{TLP} all positioned on the right side of the PC1 axis and gradually increasing, while K, Amass, LNC, LPC, and HSM were positioned on the left side of the PC1 axis and gradually decreasing, explaining 35.14% of the total variance.

PCA of hydraulic traits showed that the first two principal components together explained 74.23% of the total variance (Fig. 4d). The PC1 was represented by hydraulic safety strategy-related traits including VD and Ψ_{TLP} all positioned on the left side of the PC1 axis and gradually decreasing, while K_s , K_l , K, D_{stem} , and HSM were positioned on the right side of the PC1 axis and gradually increasing, explaining 51.46% of the total variance. PCA of economic traits showed that the first two principal components together explained 57.95% of the total variance (Fig. 4e). The PC1 was represented by acquisition strategy-related traits including LNC, LPC, WNC, and WPC, all positioned on the left side of the PC1 axis and gradually decreasing, while Amass, LMA, LT, LCC, LCN, WCC, and WCN were positioned on the right side of the PC1 axis and gradually increasing, explaining 34.72% of the total variance. In all four PCA plots, DP and RP species exhibited clear separation in their distribution: with regard to the stem and hydraulic PC1 axes, DP species were located on the left side, whereas RP species were positioned on the right; on the leaf and economic PC1 axes, DP species were situated on the right, while RP species appeared on the left. In contrast, SRP species were distributed across both sides.

In addition, we performed a linear regression analysis between the hydraulic principal component

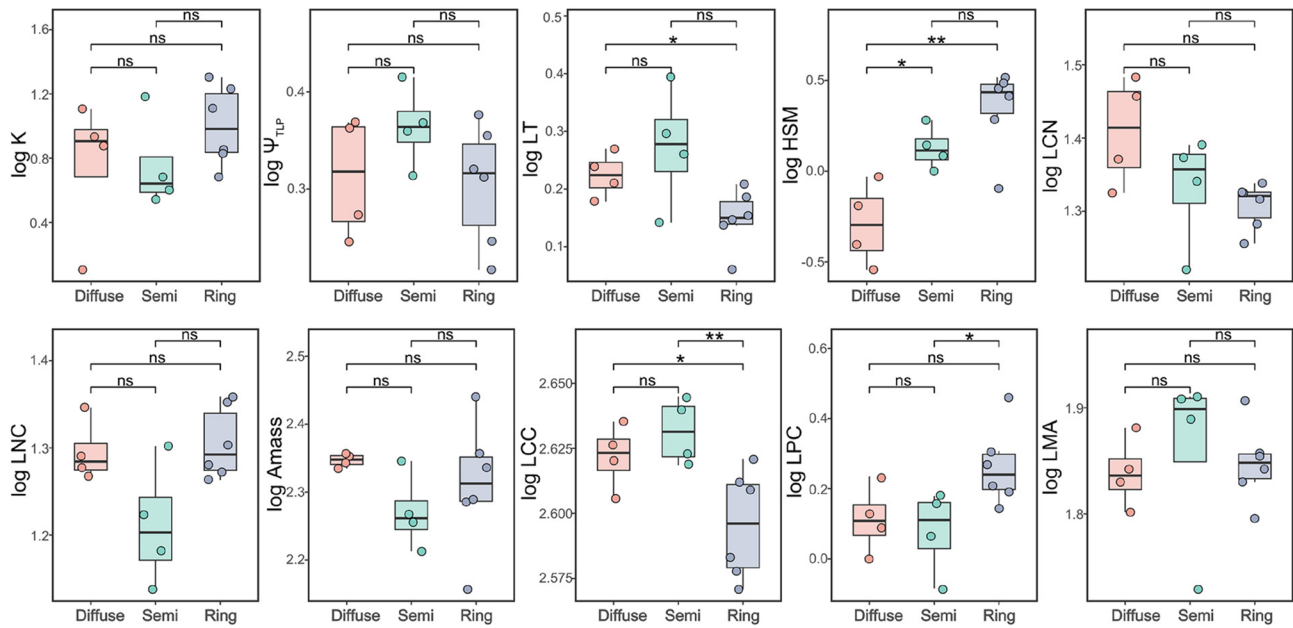


Fig. 3 Effect of urbanization on leaf functional traits. K: leaf hydraulic conductivity, Ψ_{TLP} : turgor loss point, LT: leaf thickness, HSM: hydraulic Safety Margin, LCN: leaf carbon to nitrogen ratio, LNC: leaf nitrogen content, Amass: mass-based photosynthetic rate, LCC: leaf carbon content, LPC: leaf phosphorus content, LMA: leaf mass per area. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, ns = non-significant

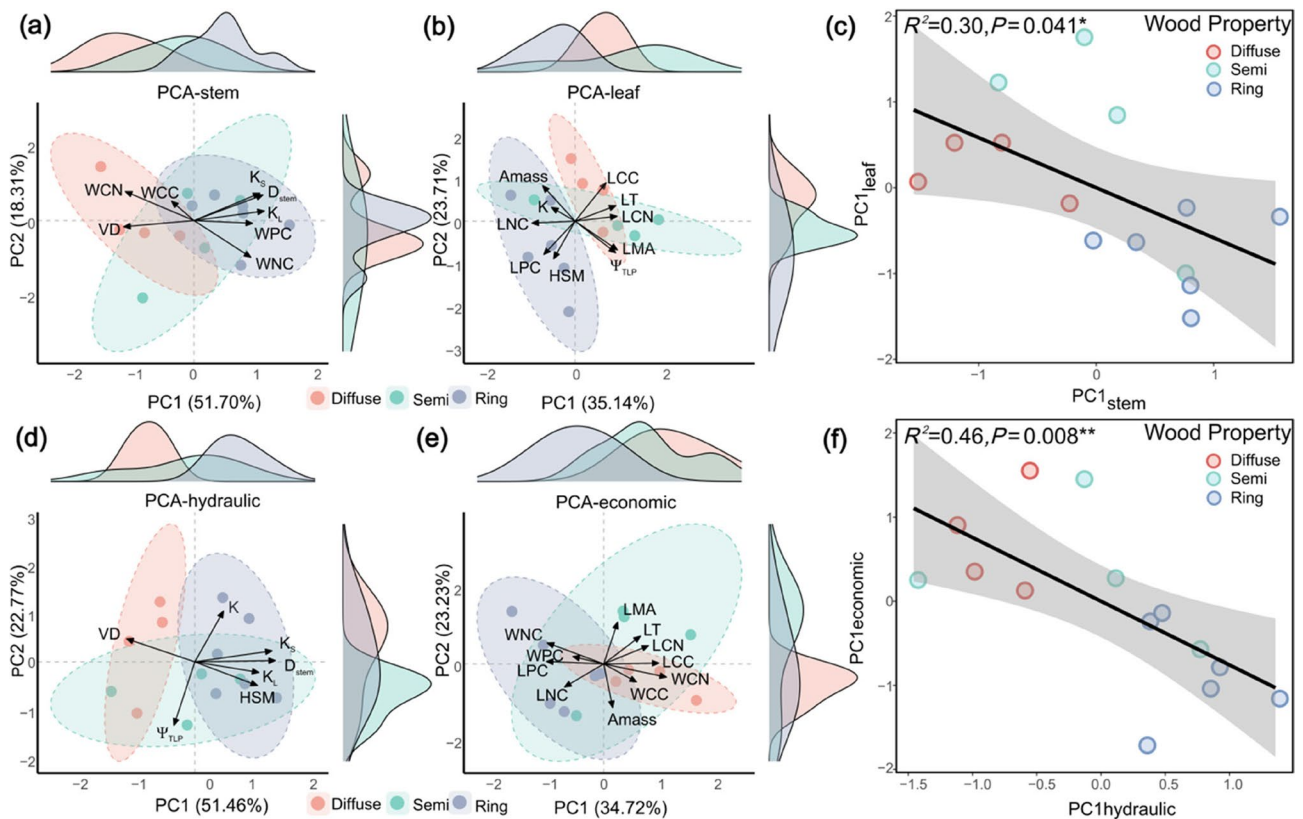


Fig. 4 Results of principal component analysis of stem (a) and leaf functional traits (b). Correlation between stem principal component PC1_{stem} and leaf principal component PC1_{leaf} (c). Principal component analysis of hydraulic traits (d) and economic traits (e). Correlation between hydraulic principal component PC1_{hydraulic} and economic principal component PC1_{economic} (f)

(PC1_{hydraulic}) and the economic principal component (PC1_{economic}). The results showed a significant negative correlation between them (Fig. 4f, $P < 0.01$). Furthermore, we examined whether the branch principal component (PC1_{stem}) is associated with the leaf principal component (PC1_{leaf}), and also found a significant negative correlation (Fig. 4c, $P < 0.05$).

For the association between branch PC1_{leaf} and functional traits, K_s showed close association with K_L , D_{stem} , and VD. Among these, K_s was positively correlated with K_L and D_{stem} but negatively correlated with VD. WCN was positioned opposite to K_L but aligned with VD and WNC (Fig. 5a). For the association between PC1_{stem} and functional traits, the results showed that Ψ_{TLP} was negatively correlated with K and Amass but positively correlated with LMA. LPC was positioned opposite to LCC but aligned with LNC. LMA correlated positively with LCN but negatively with Amass. LCN correlated negatively with HSM (Fig. 5b).

Discussion

In the present study, we systematically evaluated the differentiation and coordination of 18 functional traits across 14 urban tree species categorized by xylem anatomical structure, thereby elucidating the coupling relationships between hydraulic and economic traits and assessing adaptive strategies among species with distinct xylem anatomies. Significant disparities in both economic and hydraulic traits were observed between RP and DP species, although these differences were attenuated under urban aridity and heat stress conditions. Compared to DP species, RP species exhibited more acquisitive trait values across spectrum. As hypothesized, they were characterized by higher hydraulic conductivity (e.g., K_s , K , K_L) and increased N-P investment, constituting a “fast-return” strategy consistently manifested at both leaf and branch levels. This paper subsequently focuses on elucidating the mechanistic associations between traits related to the stem-leaf economic spectrum and hydraulic spectrum, examining interspecific variation, and interpreting how these traits underlie differential adaptation to urban drought and heat island effects among RP, DP, and SRP species groups.

Stem-leaf hydraulic functional divergence in trees with contrasting xylem anatomical types

Our study reveals a complex pattern of trait variation among xylem anatomy types under urban environmental stress. While our findings only partially support our initial hypothesis, they clearly show that not all functional differences diminished. Instead, we observed a divergence in stem hydraulic traits but a convergence in key leaf economic traits between RP and DP species. The significant differences ($p < 0.05$) in stem hydraulic

architecture (e.g., K_s , K_L , D_{stem} , VD) and wood nutrient content (WPC, WCN) confirm that the fundamental, anatomy-based strategies of RP and DP species remain distinct even under urban pressure, suggesting that this hydraulic-economic divergence is consistent with patterns observed in natural arid environments [52, 53]. This suggests that inherent anatomical designs strongly dictate stem hydraulic and carbon economic strategies.

Notably, several key leaf functional traits—including K , Ψ_{TLP} , Amass, LMA, LCN and LNC/WNC—exhibited no significant divergence ($p > 0.05$) between RP and DP species in our study, contrasting with prior reports [53, 54]. We posit that intense environmental filtering pressures within the shared urban habitat (e.g., chronic drought or thermal extremes) may drive convergent evolution or phenotypic plasticity in leaf functional traits across xylem anatomical types [16, 27], thereby attenuating the influence of xylem structural divergence on foliar strategies [55–57]. Empirical studies further indicate that environmental filtering exerts stronger selective pressure on the resource-conservative end of the leaf economics spectrum, characterized by traits including high LMA, reduced foliar nitrogen levels, and lower photosynthetic capacity (Fig. 4b). This selective pressure leads phylogenetically distinct lineages to converge on similar adaptive strategies in resource-constrained environments [58, 59]. This convergence implies that within the urban drought matrix examined here, DP and RP species may undergo parallel adjustments in foliar functional traits to align with habitat demands, effectively decoupling leaf-level adaptations from wood anatomical constraints.

Spectrum of efficiency-safety trade-offs and hydraulic-economic coordination

Our findings revealed that RP and DP species employ divergent hydraulic safety strategies, which deviates from the classical unidimensional efficiency-safety paradigm [60, 61]. Specifically, PCA of hydraulic traits (Fig. 4d) delineated two distinct strategic pathways: RP species adopt a “high-efficiency, high-margin” strategy, prioritizing rapid water transport and relying on a wide hydraulic safety margin (HSM) for protection. In contrast, DP species exhibit a “construction-oriented safety” strategy, investing in high VD and more Ψ_{TLP} to enhance structural resistance to embolism. As anticipated, SRP species exhibited an intermediate strategy, which was confirmed by our results for both hydraulic spectra and stem-leaf traits, and reflecting their transitional xylem anatomy. This strategic divergence was underpinned by stark contrasts in xylem anatomy and nutrient investment. The RP species formed larger vessel diameter and lower vessel density ($P < 0.001$), followed by SRP species (density lower than DP) (Fig. 2). This explains the higher K_s and K_L of RP species compared to DP species. In addition, the

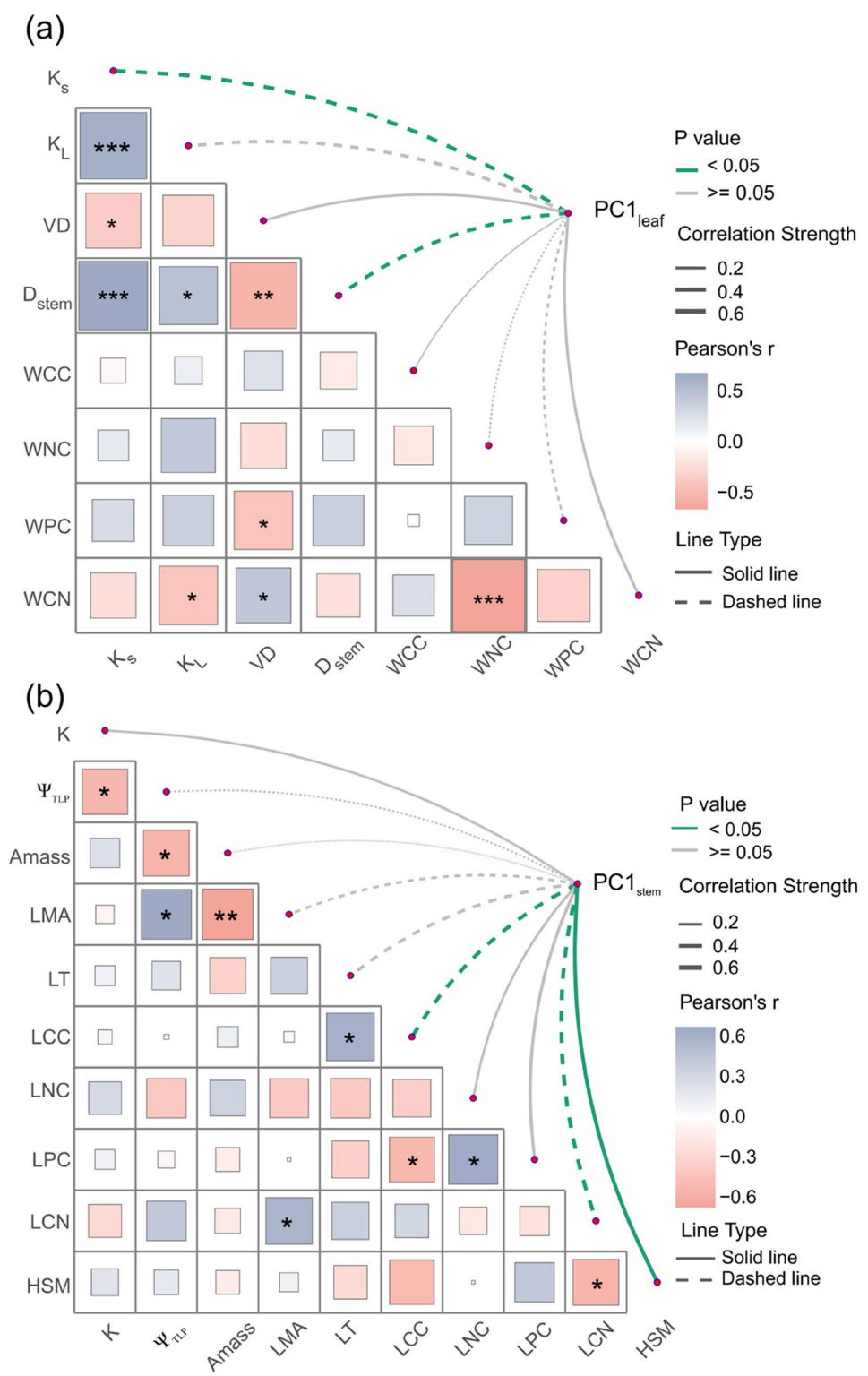


Fig. 5 Heat map of branch (a) and leaf functional traits (b). The depth of the lattice color represents the strength of the correlation, and the number of asterisks represents statistical significance. Green line segments indicate significant correlation, gray line segments are non-significant, solid lines indicate positive correlation, and dashed lines indicate negative correlation. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

wider vessels reduce hydraulic resistance to water transport, primarily due to the inversely proportional relationship between frictional resistance and the fourth power of vessel radius according to Poiseuille's law [62]. In RP trees, these wider vessels were associated with lower drought resistance, structural vulnerability and heightened embolism susceptibility [63, 64]. Furthermore, stem differentiation analysis showed significantly lower WCN in RP species compared to DP species, confirming their reduced structural defense investment. Concurrently, the elevated phosphorus content in RP species corresponds to the heightened frequency requirement for embolism repair in branches [65, 66]. Moreover, the higher HSM in RP species may partially compensate for their reduced structural investment in branches, a result that diverges from prior reports [35]. This discrepancy could stem from hydraulic safety enhancement mechanisms in some RP taxa, including dynamic embolism repair [8] and xylem osmotic adjustment [67, 68]. SRP species may similarly leverage such mechanisms, as their transitional xylem architecture could confer superior hydraulic resilience compared to the homogeneous DP wood. Thus, our results confirmed that within the efficiency-safety trade-off framework, RP species exhibit a “risk-prone strategy” whereas DP species display a “risk-averse strategy”, with RP species demonstrating greater water transport efficiency, however, this high efficiency comes at the cost of reduced drought adaptability compared to the more conservative DP species. SRP species occupy an intermediate position between RP and DP groups, filling the adaptive strategy space between these extremes. This strategy may grant SRP species greater adaptive capacity to cope with diverse urban environmental challenges [69].

Beyond significant divergence in hydraulic traits, the two wood types exhibited divergence in traits associated [28, 70] with the leaf hydraulic-economics spectrum (Fig. 4b) and whole-plant economics spectrum (Fig. 4e), revealing contrasting evolutionary strategies for resource investment and drought adaptation. Our findings demonstrate that RP species exhibited a “resource-acquisitive” strategy characterized by thinner leaves, lower LCC/WCC, and higher investment in nitrogen (LNC/WNC) and phosphorus (LPC/WPC). This suite of traits reflects a carbon economy that prioritizes rapid returns on investment: reduced structural carbon costs enable higher photosynthetic efficiency per unit mass, while the elevated nitrogen and phosphorus contents we observed (Fig. 1) support high metabolic activity and potential embolism repair mechanisms [65, 66]. The combination of thin leaves and higher HSM represents an economic strategy in which carbon is allocated toward physiological resilience rather than structural durability. In contrast, DP species displayed a “resource-conservative” strategy with thicker leaves and higher LCC/WCC and

LCN/WCN. We propose that this reflects a substantial initial carbon investment in structural defenses (e.g., thickened cuticles and cell walls) [71, 72], resulting in extended leaf lifespans and reduced resource loss under stress. This structural reinforcement is consistent with their conservative hydraulic strategy and indicates an economic priority for damage avoidance and tissue protection over rapid acquisition. SRP species exhibited an intermediate position along both spectrums, particularly evident in their leaf traits. Their carbon investment levels surpassed those of RP species, yet they maintained intermediate HSM values, suggesting a distinct economic trade-off. This pattern may indicate either: less efficient carbon use for structural purposes, or investment in different types of structural compounds that provide alternative protective functions [73, 74]. However, such compensatory mechanisms are not universally effective, as their operational success remains contingent upon species-specific anatomical adaptations and environmental constraints [75, 76].

Through further analysis of the hydraulic-economic spectrum, we found that the significant negative correlation between PC1hydraulic and PC1economic (Fig. 4f) demonstrates a fundamental coordination and reveals a tight coupling between hydraulic and economic strategies: species with an acquisitive hydraulic strategy (high efficiency, high safety margin) also employ an acquisitive economic strategy (high nutrient content, high photosynthetic rate). Conversely, species with a conservative hydraulic strategy (high vessel density, resistant to embolism) are aligned with a conservative economic strategy (high carbon investment, structural reinforcement). This coordinated variation across spectra underscores that hydraulic and economic functions have evolved synergistically to form integrated whole-plant strategies in response to environmental pressures [77–79].

Stem and leaf level mechanisms of hydraulic-economic coupling

We expected that significant synergistic patterns in functional traits between stems and leaves, highlighting the pivotal role of multi-organ integration in resource allocation and adaptive strategies. Specifically, the negative correlation (Fig. 4c, $p < 0.05$) between PC1 of stem and leaf economic spectra reflects a pronounced functional coupling at the stem and leaf levels: stem investment toward high hydraulic efficiency was accompanied by leaf shifts toward acquisitive economic spectra. These findings refine the classical assumption of intra-organ trait coordination in traditional economics spectrum theory [80], further supporting the functional modularity hypothesis [81]—where plants achieve differential optimization of carbon acquisition and water transport through modular coupling between stem and leaf functional units. This

stem-leaf synergistic strategy may reflect an evolutionary integration under environmental stress between “high hydraulic efficiency-low structural investment” and “high photosynthetic gain-low stress resistance”, as predicted by the plant economics spectrum framework [82, 83].

Further analysis of stem-leaf functional traits revealed a profound regulatory role of chemical stoichiometry over anatomical architecture in resource allocation strategies. WPC content exhibited a negative correlation with VD, suggesting that P limitation—as a critical component for nucleic acid synthesis—may constrain meristematic activity, thereby reducing vessel production [84]. Conversely, WCN showed a positive correlation with VD, indicating preferential carbon allocation toward vessel wall thickening rather than nitrogen-dependent metabolic functions. This anatomical-chemical coupling mechanism further elucidates the significant correlation between $PC1_{\text{stem}}$ and LCN: high carbon investment in stem structural reinforcement is coupled with reduced nitrogen allocation to leaves, prompting foliar tissues to maintain cell wall integrity through elevated C: N ratios, which consequently narrowing hydraulic safety margins (negative correlation with LCN, $p < 0.05$). These findings extend the classical “efficiency-safety” trade-off paradigm [85, 86] by highlighting the regulatory hierarchy of mineral nutrients (N, P) in xylem development—where stoichiometric constraints mediate anatomical optimization under competing physiological demands.

Collectively, our findings provide strong evidence for a coherent stem-leaf spectrum of plant economic strategy. The cross-scale coordination of stem-leaf traits underscores systemic integration costs in plant adaptive strategies. Consistent with the “fast-conservative” axis of the classical leaf economics spectrum, A_{mass} were negatively correlated with LMA. Simultaneously, the significant association between stem hydraulic conductivity and $PC1_{\text{leaf}}$ suggests direct hydraulic modulation of photosynthetic strategies: high-conductivity stems may transiently support elevated stomatal conductance in leaves, yet necessitate Ψ_{TLP} downregulation (negative correlation with A_{mass} , $p < 0.05$) to avoid hydraulic failure. While this dynamic equilibrium mitigates short-term water stress, its efficacy is constrained by stoichiometric rigidities (e.g., C: N ratio thresholds), which limit long-term stress resilience and amplify adaptive risks under fluctuating environments—a phenomenon exacerbated by climate-driven hydraulic volatility [24, 47]. Crucially, these antagonistic and synergistic stem-leaf trait relationships are not merely products of environmental filtering, but inevitable outcomes of multilevel optimization spanning anatomical, physiological, and stoichiometric dimensions to maximize fitness. This integrative framework advances our capacity to decode plant adaptation in

heterogeneous habitats, bridging organ-scale trade-offs with whole-plant evolutionary strategies.

Conclusion

Our findings imply that RP and DP species represent two opposing ends of the acquisitive-conservative spectrum in urban arid environments, while SRP species occupy a transitional position along this continuum. Despite sharing habitats, tree species with these xylem anatomical types exhibit diametrically opposed functional profiles, with pronounced contrasts in stem hydraulic conductance (e.g., K_S , K_L) and leaf economic traits (e.g., WCN, LCC). RP species exhibit a hydraulic safety enhancement strategy, demonstrating wider HSM and higher LPC/WPC than DP species. The coordinated patterns of stem-leaf functional modules reveal intrinsic mechanisms of hydraulic-safety trade-offs mediated through inter-organ coordination. These findings further provide evidence that the fast-slow spectrum underlies the hydraulic-economic trade-off in urban drought adaptation strategies among RP, SRP, and DP species. Nevertheless, manipulative experiments are required to investigate the eco-physiological mechanisms governing such multidimensional trade-offs.

Abbreviations

DP	Diffuse-porous
RP	Ring-porous
SRP	Semi-ring-porous
K_S	Xylem-specific conductivity
K_L	Leaf-specific conductivity
D_{stem}	Stem xylem vessel diameter
VD	Vessel density
WCN	Wood carbon to nitrogen ratio
WPC	Wood phosphorus content
WCC	Wood carbon content
WNC	Wood nitrogen content
K	Leaf hydraulic conductivity
Ψ_{TLP}	Turgor loss point
LT	Leaf thickness
HSM	Hydraulic Safety Margin
LCN	Leaf carbon to nitrogen ratio
LNC	Leaf nitrogen content
A_{mass}	Massbased photosynthetic rate
LCC	Leaf carbon content
LPC	Leaf phosphorus content
LMA	Leaf mass per area

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

YL conceived this study. NH, LL conducted the experiment. SL wrote the main manuscript. YM and JW supervised the study. All authors have read and agreed to the definitive version of the manuscript.

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Data availability

Datasets are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This experiment does not involve human experiments and animal experiments. The field experiments in the current study were permitted by the local government in China (Agriculture, Forestry and Water resources Bureau of Luyang District, Hefei City), including the collection of leaf and stem samples. The first author identified and deposited voucher specimens (NO. AAU20230801, NO. AAU20230802) in the herbarium of School of Forestry and Landscape Architecture, Anhui Agricultural University. The authors confirm that all methods comply with relevant institutional, national, and international guidelines and legislation for plant ethics.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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